

Connecting TCP/IP Instruments to qmeas

Static addressing, switch wiring, verification, and device entry

Read this first. Work strictly in the order below. Every step verifies the one before it. If you add the device to qmeas before **ping** succeeds, a failure tells you nothing: you cannot distinguish a cabling fault from a wrong subnet, a busy instrument, or a bad command string. Ninety percent of "qmeas cannot reach the instrument" cases are resolved before qmeas is ever opened.

0. What you need, and how the network is meant to look

The usual measurement setup is an **isolated instrument LAN**: one unmanaged switch, the measurement PC, and the instruments. No DHCP server exists on that switch, which is exactly why nothing gets an address on its own and why every participant must be given a fixed one by hand.

- An unmanaged (dumb) switch and one patch cable per participant. Any modern gigabit switch will do; no configuration is needed or possible.
- The measurement PC, ideally with a **second, dedicated network adapter** for instruments, keeping the institute network on the first.
- Each instrument's front panel or web interface, to set its address.
- Ten minutes and a written plan of which address belongs to which box.

A switch needs no crossover cable and no special ports; modern hardware auto-negotiates. If you are connecting exactly one instrument and nothing else, you can skip the switch and use a single cable directly between PC and instrument. Everything else in this document is identical either way.

Common trap with a switch. A switch forwards traffic but assigns nothing. Devices left on "DHCP" on an isolated switch fall back to a random 169.254.x.x self-assigned address and will not talk to anything reliably. If you see 169.254 anywhere, that device never got an address - it did not "partly work".

1. Plan the subnet before touching anything

Every device on the switch - the PC included - must sit in the **same subnet**. With the usual mask 255.255.255.0, that means the first three numbers are identical for everyone and only the last differs. Write the plan down before configuring; it takes a minute and prevents most of the confusion later.

Role	Address	Mask	Note
Measurement PC	192.168.1.100	255.255.255.0	Pick something clear of the instruments, e.g. .100
Keithley 2450	192.168.1.1	255.255.255.0	VISA / INSTR
SR860 lock-in	192.168.1.2	255.255.255.0	VISA / INSTR
Magnet controller	192.168.1.3	255.255.255.0	Raw socket, port 7020
OptiCool	192.168.1.30	255.255.255.0	Raw socket, port 5100

These are the addresses used in the example device lists shipped with qmeas (devices/opticool/, devices/proteox/). Yours may differ - what matters is that the first three numbers agree across the whole switch.

The gateway: leave it empty

A gateway is the router used to reach networks *other* than your own. An isolated instrument LAN has no router and needs to reach nothing else, so **leave the gateway field blank**. This is a deliberate correction to the common instinct of filling in every field.

It matters concretely: if the PC has a second adapter for the institute network, entering a gateway on the *instrument* adapter creates a second default route. Windows may then send internet traffic into a switch that goes nowhere, and you lose network access for reasons that look completely unrelated to the instrument you were setting up. Leave it blank and that class of problem cannot occur.

If a field refuses to stay empty, or a device demands a gateway, entering its own address or the .1 of the subnet is harmless on an isolated LAN - just never do so on the adapter that also carries your internet connection.

Choosing the address range

Use a private range: 192.168.x.x is conventional and fine. Two constraints only: do not reuse the range your institute network already uses on the other adapter (Windows cannot decide which adapter to use, and both will behave erratically), and give every device a distinct final number.

Two instruments, two different subnets. A real example from the shipped device lists: the kiutra controller sits at 192.168.11.20 while everything else is on 192.168.1.x. Those are different subnets and will not see each other with a /24 mask. Three ways out: change the instrument to match, widen the PC mask to 255.255.0.0 so it spans both, or give the PC a second address on the same adapter (Advanced button in the dialog in step 2). The first is cleanest when the instrument allows it.

2. Give the Windows adapter a fixed address

Windows defaults to "Obtain an IP address automatically", which waits for a DHCP server that does not exist on your switch. Change it to a fixed address.

Through the GUI

#	Do this
1	Press Win+R , type <code>ncpa.cpl</code> , press Enter. This opens Network Connections directly and skips several layers of Settings menus.
2	Identify the adapter connected to the instrument switch. If several look alike, unplug the cable for a moment - the one that reports "Network cable unplugged" is the right one.
3	Right-click it, choose Properties . Consider renaming the adapter now (e.g. "Instruments") - it makes the command-line form below far easier.
4	Select Internet Protocol Version 4 (TCP/IPv4) in the list, then click Properties . Select the row itself; do not untick its checkbox.
5	Switch from "Obtain an IP address automatically" to Use the following IP address .
6	Enter the address and mask from your plan. Leave the gateway blank; leave the DNS fields blank.
7	Confirm with OK , then OK again in the adapter properties. The second dialog must also be closed for the setting to apply.

IP address: 192.168.1.100
Subnet mask: 255.255.255.0
Default gateway: (leave empty)

Or from the command line

Faster, scriptable, and unambiguous about which adapter you mean. Run **cmd as Administrator**. Substitute your adapter name exactly as it appears in Network Connections:

```
netsh interface ip set address name="Instruments" static 192.168.1.100 255.255.255.0
:: put it back to automatic later:
netsh interface ip set address name="Instruments" dhcp
```

Then confirm what actually took effect:

```
ipconfig
Ethernet adapter Instruments:
IPv4 Address. . . . . : 192.168.1.100
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

An address starting 169.254 here means the static setting did not apply - most often the second OK was never clicked, or the change was made on the wrong adapter.

3. Set the address on the instrument

Every instrument needs the same treatment: static (not DHCP), an address from your plan, and the matching mask. Menu paths differ by manufacturer and firmware, so treat the following as orientation and confirm against the instrument's own manual.

Instrument	Where to look	Watch out for
Keithley 2450	Front panel MENU key, then the Communication / LAN settings. Set the TCP/IP configuration mode to manual, enter address and mask.	Some firmware requires a LAN reset or a power cycle before the new address is live. If it will not take, power-cycle before assuming failure.
SR860 lock-in	Front panel setup / configuration menu, Ethernet section. Switch from DHCP to static and enter the address.	The display shows the currently active address - use it to confirm the setting took, rather than trusting the entry screen.
OptiCool	Configured in the Quantum Design software / controller rather than a front panel. The instrument itself listens on a raw socket port (5100 in the shipped example).	The control software may need restarting after a change. Note the port as well as the address; you need both later.
Magnet / cryostat controller	Usually a web interface or vendor tool rather than a front panel. Consult its manual for the LAN page.	Frequently accepts only one client at a time - see the note in step 5.

Write down the port, not only the address. Instruments using raw sockets (OptiCool 5100, magnet 7020) or HTTP APIs need the port number in the qmeas address string. VISA/INSTR devices do not. Noting it now saves a trip back to the instrument later.

4. Verify with ping before opening qmeas

This is the step that is worth the discipline. **ping** tests exactly one thing - whether the PC and the instrument can exchange packets - and it does so without any involvement from qmeas, VISA, or command syntax. Open **cmd** and ping the instrument:

```
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
```

```

Reply from 192.168.1.2: bytes=32 time<1ms TTL=64
Reply from 192.168.1.2: bytes=32 time<1ms TTL=64
Reply from 192.168.1.2: bytes=32 time<1ms TTL=64
Reply from 192.168.1.2: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

```

Four replies, zero loss, sub-millisecond times on a local switch. This is what success looks like.

When it fails, the wording tells you where to look

Response	What it means	What to check
Destination host unreachable	Your PC has no route to that subnet.	PC and instrument are in different subnets, or the PC adapter never took its static address. Re-run ipconfig.
Request timed out	The route exists; nothing answered.	Instrument off, still on DHCP, on a different address than you think, cable in the wrong port, or a firewall blocking ICMP.
General failure	The local adapter is not usable.	Cable unplugged, adapter disabled, or no link light on the switch.
Some replies, some lost	A real physical problem.	Bad cable or a failing switch port. Swap the cable first; it is the cheapest test and often the answer.

Ping succeeding is necessary, not sufficient

ICMP (what ping uses) and the instrument's command port are different things. A device can answer ping perfectly while its SCPI port is closed, occupied by another program, or simply a different number than you assumed. Test the actual port from PowerShell:

```

Test-NetConnection -ComputerName 192.168.1.30 -Port 5100

ComputerName : 192.168.1.30
RemotePort   : 5100
TcpTestSucceeded : True

```

`TcpTestSucceeded : False` with a successful ping is a precise and useful result: the network is fine and the problem is the port, the service, or another client holding the connection.

Do not skip ahead. If ping does not reply, nothing in qmeas can help and no address string will fix it. Stay in cmd until it answers.

5. Add the device to qmeas

Only now open qmeas. In **Devices & Commands**, create or edit the device and enter its address. The syntax depends on how the instrument speaks, and this is the field most often entered wrongly:

Instrument type	Address form in qmeas	Example
VISA instrument (most SCPI boxes)	TCPIP0:: <ip>::inst0::INSTR</ip>	TCPIP0::192.168.1.1::inst0::INSTR
VISA, short form	TCPIP0:: <ip>::INSTR</ip>	TCPIP0::192.168.1.2::INSTR
Raw socket (port required)	TCPIP0:: <ip>::<port>::SOCKET</ip>	TCPIP0::192.168.1.30::5100::SOCKET

Instrument type	Address form in qmeas	Example
HTTP / JSON-RPC API	HTTP:: <ip>::<port>::<path>::<content-type></ip>	HTTP::192.168.11.20::1006::application/json-rpc

Getting this wrong produces misleading symptoms. A raw-socket instrument addressed as ::INSTR typically times out rather than reporting a syntax problem, which looks exactly like a network fault you have already ruled out.

Terminators and timeout

Alongside the address, the device settings carry a write terminator, a read terminator, and a timeout in milliseconds. The terminator choices are none, \r, \n, \r\n and \n\r; most SCPI instruments expect \n on both, which is the default in the shipped examples, with 5000 ms as a sane timeout.

The read terminator matters more on raw sockets than people expect: qmeas reads until it sees that terminator, so a wrong setting can truncate a reply mid-transfer and leave data unread in the receive buffer. On instruments that accept a single client, that is a known route to intermittent connection resets.

Test Connection

Use **Test Connection** in the device editor as the final check. For VISA and socket devices it sends *IDN? and shows the reply - identifying the box by name, which also confirms you are talking to the instrument you think you are. For HTTP devices there is no universal identity query, so it tests TCP reachability of host and port only; use **Query Once** on a real command to exercise the API itself.

One client at a time. Many instrument servers accept a single connection. If NI MAX, a vendor tool, a browser tab, or a second qmeas instance holds it, qmeas gets a refused or reset connection while ping still succeeds perfectly. Close other software before concluding something is broken.

6. Troubleshooting, in the order worth trying

Symptom	Most likely cause and remedy
Ping fails, PC shows 169.254.x.x	The static address never applied. Re-open the IPv4 properties and confirm both dialogs were closed with OK, or set it with netsh and verify with ipconfig.
Ping fails, addresses look right	Compare the first three numbers of PC and instrument - a mismatch there is the single most common cause. Then check link lights at both ends of the cable.
Ping works, Test Connection fails	Wrong address form (INSTR vs SOCKET), wrong port, wrong terminator, or another program holding the single allowed connection.
Worked yesterday, not today	Someone moved a cable, or the instrument was power-cycled and reverted to DHCP. Check the instrument's displayed address first; it is faster than re-deriving the setup.
Internet stopped working after setup	A gateway was entered on the instrument adapter. Clear it - see section 1.
Intermittent drops mid-measurement	Suspect the read terminator and the single-client behaviour before suspecting the cable, particularly on raw-socket devices. Also check that no vendor tool polls the same instrument in parallel.
Instrument invisible, address unknown	Ping the broadcast address, then run arp -a to list what answered on the subnet. Vendor discovery tools and NI MAX can also find devices whose address you have lost.

7. Quick reference

```
:: open network connections
ncpa.cpl

:: set / revert a static address (Administrator cmd)
netsh interface ip set address name="Instruments" static 192.168.1.100 255.255.255.0
netsh interface ip set address name="Instruments" dhcp

:: verify the PC
ipconfig

:: verify the instrument
ping 192.168.1.2

:: verify the port (PowerShell)
Test-NetConnection -ComputerName 192.168.1.30 -Port 5100

:: see who answered on the subnet
arp -a
```

Order of operations

Step	Done when
1. Plan the subnet	Every device has a distinct address sharing the first three numbers.
2. Configure the PC adapter	ipconfig shows the static address, empty gateway.
3. Configure the instrument	The instrument displays the intended address, not a DHCP one.
4. Ping	Four replies, zero loss.
5. Add to qmeas	Test Connection returns the instrument identity.